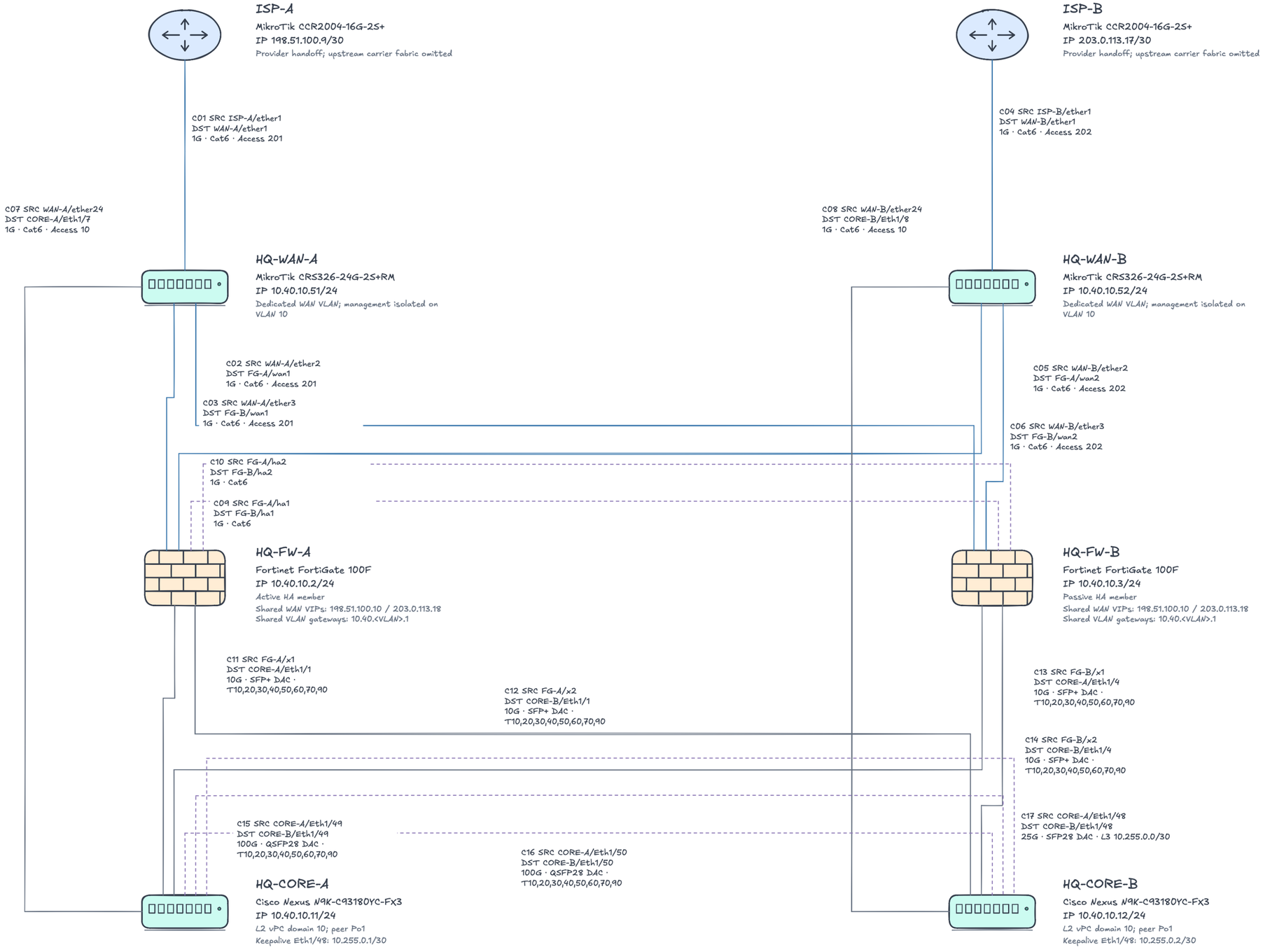
AURORA / HQ perimeter & core

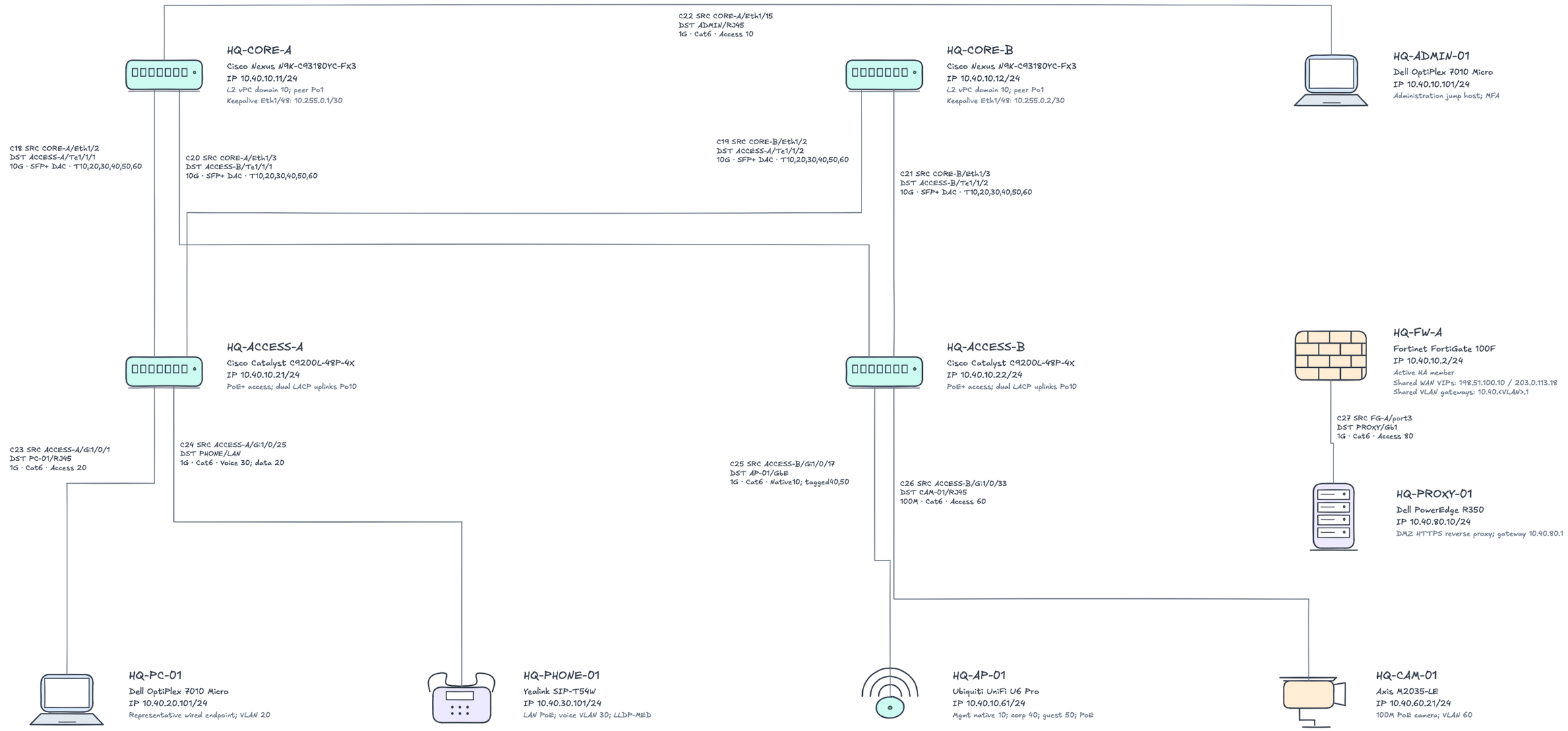
Real example hardware · fictional addressing · WAN VIPs on both HA members · vPC core · local cables only



WAN-A VLAN201 and WAN-B VLAN202 carry public handoffs only; VLAN10 administration is isolated. HA uses ha1 + ha2; core Po1 uses two 100G links.

AURORA / HQ access & DMZ

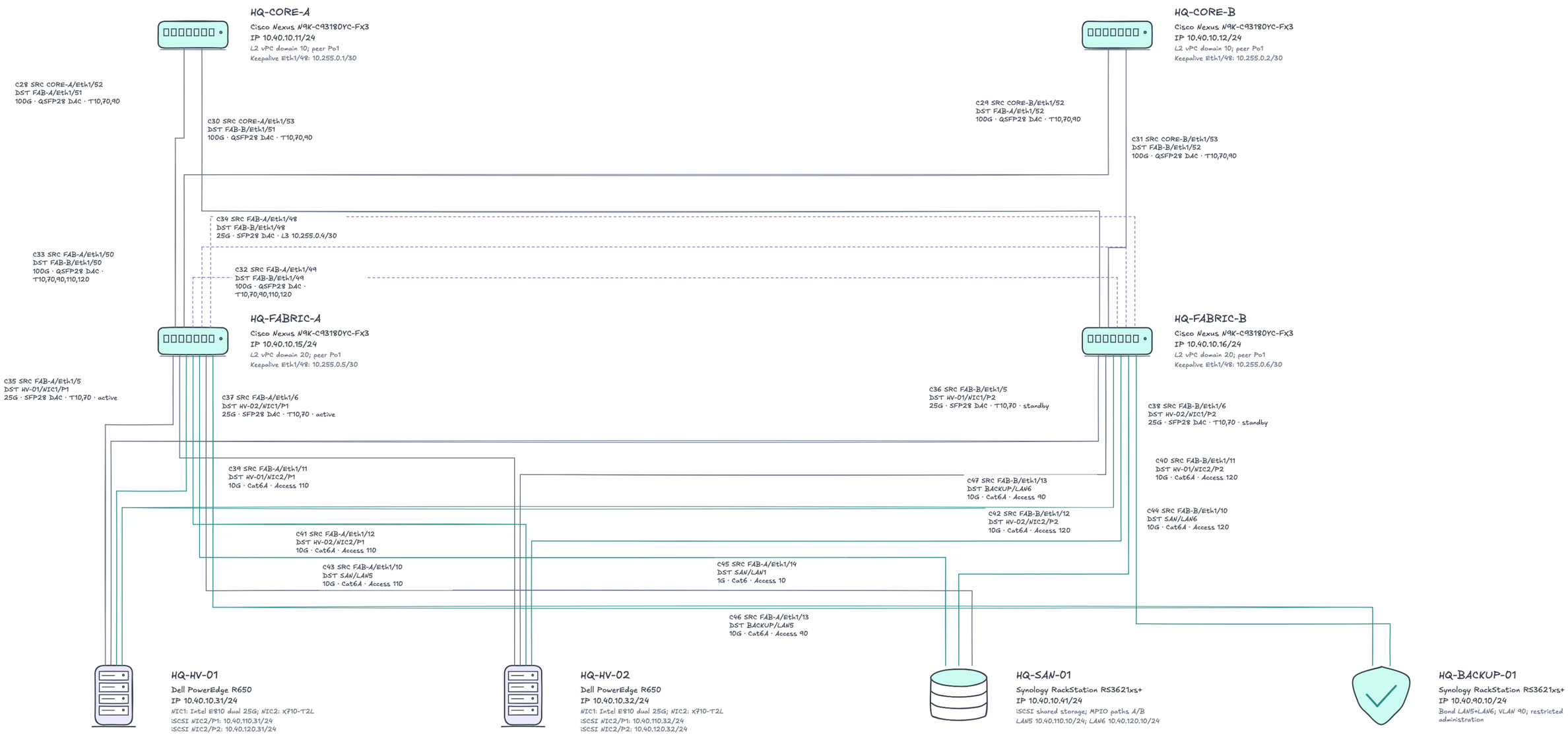
Every displayed endpoint has an exact model and IP · PoE and VLAN intent accompany each physical port label



Fictional reference deployment · every cable has SRC/DST ports · repeat symbols refer to the same inventoried device

AURORA / HQ compute & storage

Dual 100G fabric uplinks · 25G host data · two isolated 10G iSCSI paths · exact physical NIC ports



NIC1: vSwitch0 active/standby independent trunks, no LACP. NIC2: VLAN110/120 iSCSI MPIO, no gateway. Backup LAN5+LAN6: separate LACP bond.

AURORA / Québec distribution

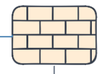
Exact WAN, gateway, switch and endpoint ports · warehouse Wi-Fi is explicitly a logical link

OPERATIONS / branch
U = untagged/native; T = tagged VLAN
WAN service 300/100 Mbps; copper link 1G
Internet ingress: no published service
IPsec to HQ; no public management
See BR-OPS for interfaces, SSIDs and ports
Phone PC passthrough is unused
All MACs, locations and allocations fictional



BR-ISP

MikroTik CCR2004-16G-2S+
IP 192.0.2.9/30
Provider handoff; carrier path beyond this scope
WAN-BR-A / FIC-BRA-003
Business fibre: 300 down / 100 up Mbps
Location: Provider demarc / BR-R01-U38
MAC: 02:50:00:00:00:01

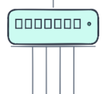


BR-FW

Fortinet FortiGate 60F
IP 10.50.10.1/24
VLAN routing + DHCP; routed IPsec to HQ
wan1 192.0.2.10/30; GW .9
internal: U10 T20,30,40,50,60
Location: Comms closet / BR-R01-U37
MAC: 02:50:00:00:00:02
Manage: https://10.50.10.1

C48 SRC ISP-BR/ether1 · L3
DST FW-BR/wan1 · L3
1G · Cat6 · Routed 192.0.2.8/30

C49 SRC FW-BR/internal1 · U10 T20,30,40,50,60
DST SW-BR/Gi1/0/48 · U10 T20,30,40,50,60
1G · Cat6 · Native10; tagged20,30,40,50,60



BR-ACCESS-01

Cisco Catalyst C9200L-48P-4X
IP 10.50.10.11/24
PoE+ access; 1G gateway trunk
Location: Comms closet / BR-R01-U34
MAC: 02:50:00:00:00:03
Manage: https://10.50.10.11

C50 SRC SW-BR/Gi1/0/1 · U20
DST PC-BR/RJ45 · U20
1G · Cat6 · Access 20

C51 SRC SW-BR/Gi1/0/17 · U10 T40,50,60
DST AP-BR/GbE · U10 T40,50,60
1G · Cat6 · Native10; tagged40,50,60 · PoE+ from SW-BR

C52 SRC SW-BR/Gi1/0/25 · U20 T30
DST PHONE-BR/LAN · U20 T30
1G · Cat6 · Voice30; data20 · PoE from SW-BR

C53 SRC SW-BR/Gi1/0/33 · U60
DST PRINT-BR/RJ45 · U60
1G · Cat6 · Access 60



BR-SCANNER-01

Zebra TC52
IP 10.50.60.61/24
Wi-Fi scanner; Aurora-IoT; no physical cable
Location: Warehouse mobile fleet; example unit
MAC: 02:50:00:00:00:08

L01 SRC AP-BR/Aurora-IoT
DST SCAN-BR/wlan0
Wi-Fi: 5GHz · SSID Aurora-IoT / VLAN60



BR-AP-01

Ubiquiti UniFi U6 Pro
IP 10.50.10.41/24
SSIDs: corp 40; guest 50; Aurora-IoT 60
Location: Warehouse aisle A / ceiling
MAC: 02:50:00:00:00:04
Manage: https://10.50.10.41



BR-PC-01

Dell OptiPlex 7010 Micro
IP 10.50.20.101/24
Dispatch endpoint; VLAN 20
Location: Dispatch desk D01
MAC: 02:50:00:00:00:05



BR-PHONE-01

Yealink SIP-T54W
IP 10.50.30.101/24
LAN PoE; voice VLAN 30
Location: Dispatch desk D01
MAC: 02:50:00:00:00:06



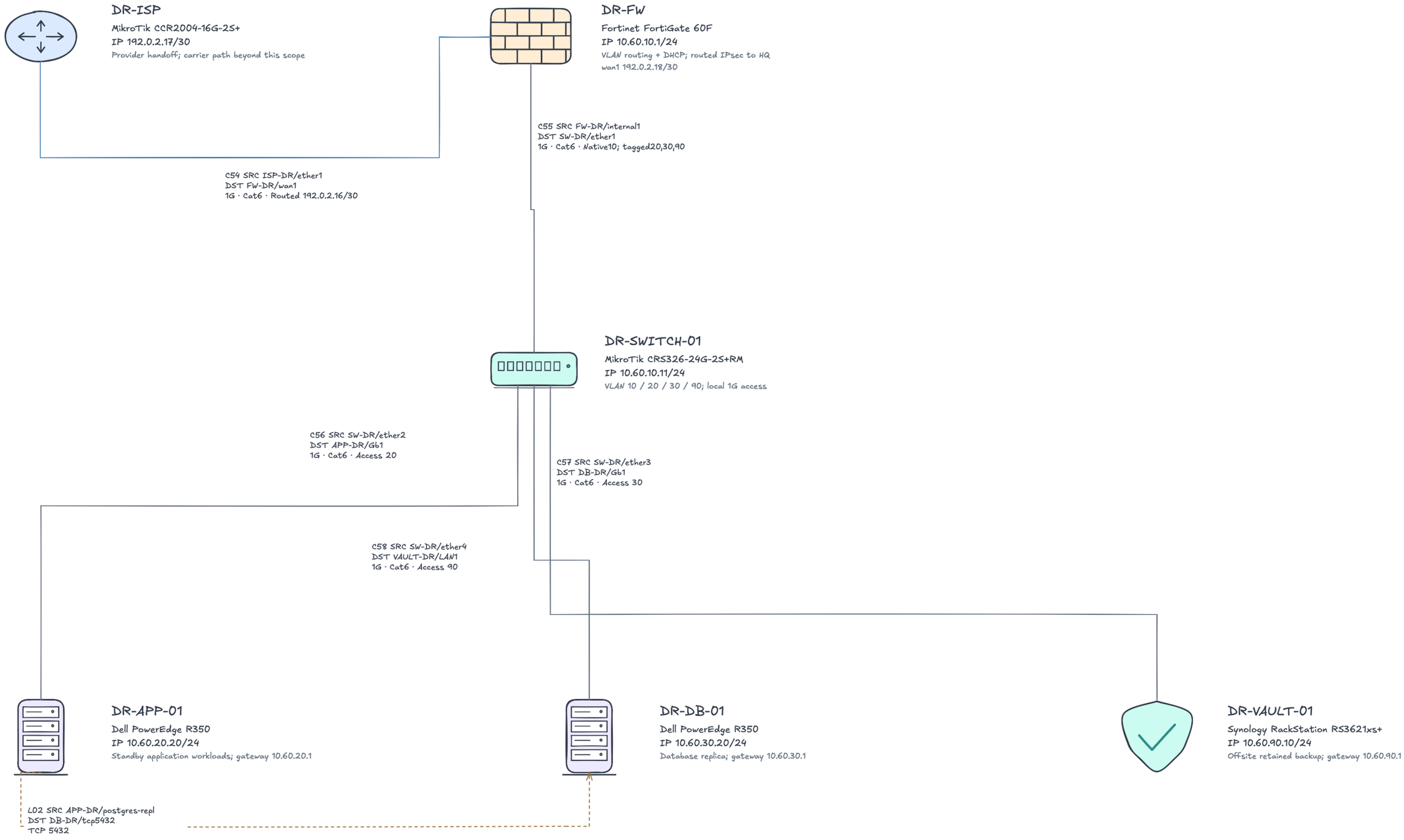
BR-PRINTER-01

Brother HL-L5210DN
IP 10.50.60.21/24
Wired label/dispatch printer; VLAN 60
Location: Dispatch bench P01
MAC: 02:50:00:00:00:07
Manage: https://10.50.60.21

Cable labels show U/T at both ends; C48 is routed L3. Native10 carries management; voice30 is tagged. Full schedules: BR-OPS.

AURORA / private DR site

Standalone *WAN* and local *VLAN* routing · application, database and retained backup have separate addresses



DR has a single gateway/switch and one *WAN*: documented remaining failure points. Database replication is routed traffic, not a direct Ethernet cable.

AURORA / VPN & replication overlay

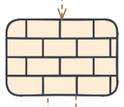
Dashed links are logical interfaces and protocols; C01–C58 on the other sheets are the physical cable plant



REMOTE-01

Dell Latitude 7450
IP 10.40.100.25/24
FortiClient IKEv2; MFA; VPN-assigned address
Outer endpoint 203.0.113.66 (fictional)

L05 SRC REMOTE/ipsec-hq
DST FG-A/ipsec-remote
IPsec IKEv2 / MFA · VPN pool 10.40.100.0/24



HQ-FW-A

Fortinet FortiGate 100F
IP 10.40.10.2/24
Active HA member
Shared WAN VIPs: 198.51.100.10 / 203.0.113.18
Shared VLAN gateways: 10.40.<VLAN>.1

L03 SRC FG-A/ipsec-br
DST FW-BR/ipsec-hq
IPsec IKEv2 · L3 10.254.0.0/30

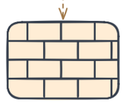
L04 SRC FG-A/ipsec-dr
DST FW-DR/ipsec-hq
IPsec IKEv2 · L3 10.254.0.4/30



HQ-BACKUP-01

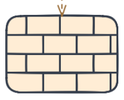
Synology RackStation RS3621xs+
IP 10.40.90.10/24
Bond LAN5+LAN6; VLAN 90; restricted administration

L06 SRC BACKUP/ssh-client
DST VAULT-DR/ssh/tcp22
SSH rsync / TCP22



BR-FW

Fortinet FortiGate 60F
IP 10.50.10.1/24
VLAN routing + DHCP; routed IPsec to HQ
wan1 192.0.2.10/30; GW .9
internal1: U10 T20,30,40,50,60



DR-FW

Fortinet FortiGate 60F
IP 10.60.10.1/24
VLAN routing + DHCP; routed IPsec to HQ
wan1 192.0.2.18/30



DR-VAULT-01

Synology RackStation RS3621xs+
IP 10.60.90.10/24
Offsite retained backup; gateway 10.60.90.1

HQ HA failover preserves the shared WAN VIPs. Remote outer IP203.0.113.66 is fictitious; the VPN pool is10.40.100.0/24. No diagram line implies a physical cable through the Internet.

AURORA / virtual services

vNIC and port-group names are logical endpoints; the physical hypervisor uplinks are C35–C38 on the compute sheet



HQ-HV-01
Dell PowerEdge R650
IP 10.40.10.31/24
NIC1: Intel E810 dual 25G; NIC2: X710-T2L
iSCSI NIC2/P1: 10.40.110.31/24
iSCSI NIC2/P2: 10.40.120.31/24

L07 SRC HV-01/vSwitch0/PG70
DST DC-01/vNIC0
Virtual switch / VMXNET3 · Port group PG70 / VLAN70

L09 SRC HV-01/vSwitch0/PG70
DST ERP-01/vNIC0
Virtual switch / VMXNET3 · Port group PG70 / VLAN70



HQ-HV-02
Dell PowerEdge R650
IP 10.40.10.32/24
NIC1: Intel E810 dual 25G; NIC2: X710-T2L
iSCSI NIC2/P1: 10.40.110.32/24
iSCSI NIC2/P2: 10.40.120.32/24

L08 SRC HV-02/vSwitch0/PG70
DST DC-02/vNIC0
Virtual switch / VMXNET3 · Port group PG70 / VLAN70

L10 SRC HV-02/vSwitch0/PG70
DST SIEM-01/vNIC0
Virtual switch / VMXNET3 · Port group PG70 / VLAN70



HQ-DC-01
vSphere virtual machine / VMXNET3
IP 10.40.70.10/24
Directory / DNS; preferred resolver
Default gateway 10.40.70.1



HQ-ERP-01
vSphere virtual machine / VMXNET3
IP 10.40.70.20/24
ERP / internal application
Default gateway 10.40.70.1



HQ-DC-02
vSphere virtual machine / VMXNET3
IP 10.40.70.11/24
Directory / DNS; secondary resolver
Default gateway 10.40.70.1



HQ-SIEM-01
vSphere virtual machine / VMXNET3
IP 10.40.70.30/24
Central logs / security monitoring
Default gateway 10.40.70.1

Both directory instances have separate IPs and host placement. Backups and DR recovery validation remain separate from VM placement.

AURORA / VLANs, gateways & deployment notes

All allocations are fictitious · HQ firewall cluster owns the .1 gateway VIPs · branch and DR have local gateways

Site	VLAN	IPv4 subnet	Gateway	Purpose	Policy / operation
HQ	10	10.40.10.0/24	10.40.10.1	Management	Admin jump host + MFA only
HQ	20	10.40.20.0/24	10.40.20.1	Wired clients	DHCP; directory/DNS + business services
HQ	30	10.40.30.0/24	10.40.30.1	Voice	LLDP-MED; PBX / SIP only
HQ	40	10.40.40.0/24	10.40.40.1	Corporate Wi-Fi	802.1X; business services
HQ	50	10.40.50.0/24	10.40.50.1	Guest Wi-Fi	Internet only; deny private destinations
HQ	60	10.40.60.0/24	10.40.60.1	IoT / cameras / printers	DNS/NTP + allowlisted services
BR	10	10.50.10.0/24	10.50.10.1	Management	Admin jump host + MFA only
BR	20	10.50.20.0/24	10.50.20.1	Wired clients	DHCP; directory/DNS + business services
BR	30	10.50.30.0/24	10.50.30.1	Voice	LLDP-MED; PBX / SIP only
BR	40	10.50.40.0/24	10.50.40.1	Corporate Wi-Fi	802.1X; business services
BR	50	10.50.50.0/24	10.50.50.1	Guest Wi-Fi	Internet only; deny private destinations
BR	60	10.50.60.0/24	10.50.60.1	IoT / cameras / printers	DNS/NTP + allowlisted services
HQ	70	10.40.70.0/24	10.40.70.1	Server workloads	Directory .10/.11; ERP .20; SIEM .30
HQ	80	10.40.80.0/24	10.40.80.1	DMZ	HTTPS reverse proxy only
HQ	90	10.40.90.0/24	10.40.90.1	Backup	Restricted admin + offsite transfer
HQ	110	10.40.110.0/24	None (L2)	iSCSI path A	L2 only; MPIO; no routed gateway
HQ	120	10.40.120.0/24	None (L2)	iSCSI path B	L2 only; MPIO; no routed gateway
DR	10	10.60.10.0/24	10.60.10.1	Management	Restricted administration
DR	20	10.60.20.0/24	10.60.20.1	Standby application	To DB TCP5432 only
DR	30	10.60.30.0/24	10.60.30.1	Database replica	Replication allowed from DR app
DR	90	10.60.90.0/24	10.60.90.1	Retained offsite backup	SSH transfer over HQ IPsec; destination retention

WAN handoffs
HQ A 198.51.100.8/30
GW .9 / HA VIP .10
HQ B 203.0.113.16/30
GW .17 / HA VIP .18
Branch 192.0.2.8/30
GW .9 / FW .10
DR 192.0.2.16/30
GW .17 / FW .18

Routed overlays
L03 HQ/BR: 10.254.0.0/30
HQ .1 / branch .2
L04 HQ/DR: 10.254.0.4/30
HQ .5 / DR .6
Remote pool: 10.40.100.0/24
HQ gateway .1; client .25

Deployment checks
1. Validate ports and optics.
2. Stage VLANs and LACP.
3. Test HA + each cable failure.
4. Verify MPIO on both paths.
5. Test guest/IoT isolation.
6. Restore a retained backup.

Known remaining failure points
Branch and DR: one firewall, one switch and one WAN each.
DMZ reverse proxy: one host.
Drawn user endpoints are representative, not a complete fleet BOM.

Port inventory, cable schedule, models and addressing are retained in the accompanying network JSON/CSV; diagrams are directly editable in Excalidraw.

AURORA / branch operations & port map

BR-OPS · interface assignments, SSID security, exact endpoint VLAN modes and PoE · generated from the inventory

WAN SERVICE / demarcation and addressing

Circuit / provider	Service / down-up	WAN / gateway	Demarcation / reference
WAN-BR-A / Aurora Fibre (fictional)	Business fibre 300 / 100 Mbps	192.0.2.10/30 GW 192.0.2.9	BR comms closet / BR-R01-U38 FIC-BRA-003; single WAN

INTERFACES / gateways and management

Device / interface	Address / assignment	VLAN / purpose
FW-BR / wan1	192.0.2.10/30 Static; GW 192.0.2.9	L3 WAN-BR-A
FW-BR / internal1	10.50.10.1/24 Static gateway	U10 Management
FW-BR / internal1.20	10.50.20.1/24 Static gateway	T20 Wired clients
FW-BR / internal1.30	10.50.30.1/24 Static gateway	T30 Voice
FW-BR / internal1.40	10.50.40.1/24 Static gateway	T40 Corporate Wi-Fi
FW-BR / internal1.50	10.50.50.1/24 Static gateway	T50 Guest Wi-Fi
FW-BR / internal1.60	10.50.60.1/24 Static gateway	T60 IoT / cameras / printers
SW-BR / VLAN10	10.50.10.11/24 Static	U10 Management / service address
AP-BR / GbE	10.50.10.41/24 DHCP reservation	U10 Management / service address
PHONE-BR / LAN	10.50.30.101/24 DHCP reservation	T30 Management / service address
PRINT-BR / RJ45	10.50.60.21/24 DHCP reservation	U60 Management / service address

WIRELESS / SSID to VLAN mapping

AP / SSID	VLAN / band	Authentication / isolation
AP-BR / Aurora-Corp	40 / 2.4 / 5 GHz	WPA2-Enterprise / 802.1X Corporate services via policy
AP-BR / Aurora-Guest	50 / 2.4 / 5 GHz	WPA2; captive portal Client isolation; Internet only
AP-BR / Aurora-IoT	60 / 5 GHz	WPA2-PSK; key stored separately Allowlisted scanner services

FIELD NOTES
DHCP: clients, voice and WLANs on BR-FW
Native management: VLAN10
Voice: tagged VLAN30 + LLDP-MED
SSID keys and login secrets kept separately
Single firewall / switch / ISP remain SPOFs
Designed facts; validate before installation

PHYSICAL PORT SCHEDULE / one row per cable

Cable	Source / VLAN	Destination / VLAN	Medium / PoE
C48	ISP-BR / ether1 L3	FW-BR / wan1 L3	1G / Cat6 No PoE
C49	FW-BR / internal1 U10 T20,30,40,50,60	SW-BR / Gi1/0/48 U10 T20,30,40,50,60	1G / Cat6 No PoE
C50	SW-BR / Gi1/0/1 U20	PC-BR / RJ45 U20	1G / Cat6 No PoE
C51	SW-BR / Gi1/0/17 U10 T40,50,60	AP-BR / GbE U10 T40,50,60	1G / Cat6 PoE+ from SW-BR
C52	SW-BR / Gi1/0/25 U20 T30	PHONE-BR / LAN U20 T30	1G / Cat6 PoE from SW-BR
C53	SW-BR / Gi1/0/33 U60	PRINT-BR / RJ45 U60	1G / Cat6 No PoE

U = untagged/native · T = tagged · L3 = routed · Service bandwidth is separate from Ethernet link speed · all facts fictional

AURORA / HQ WAN & published services

EDGE-OPS · circuit references, demarcation, upstream gateways and exact NAT translation · Fictional proposed configuration

WAN CIRCUITS / both providers and HA shared addresses

Circuit / provider	Service / down-up	WAN / gateway	Demarcation / reference
WAN-HQ-A / Aurora Fibre (fictional)	Dedicated fibre 1000 / 1000 Mbps	198.51.100.10/30 GW 198.51.100.9	HQ room M01 / R01-U42 FIC-HQA-001; shared HA VIP
WAN-HQ-B / Metro Cable (fictional)	Cable backup 500 / 50 Mbps	203.0.113.18/30 GW 203.0.113.17	HQ room M01 / R01-U41 FIC-HQB-002; shared HA VIP

DESTINATION NAT / source-restricted HTTPS portal

Rule / firewall	Ingress / source	Public service	Translated destination	Purpose / state
N01 / FG-A	wan1 / HA shared VIP 203.0.113.64/28	198.51.100.10 TCP/443	PROXY / 10.40.80.10 TCP/443	Partner HTTPS portal; log + inspect Fictional proposed rule
N02 / FG-A	wan2 / HA shared VIP 203.0.113.64/28	203.0.113.18 TCP/443	PROXY / 10.40.80.10 TCP/443	Same portal via backup ISP Fictional proposed rule

INTERFACE / ADDRESS ROLES
wan1: HA VIP 198.51.100.10/30; GW .9
wan2: HA VIP 203.0.113.18/30; GW .17
Unique management: FG-A .10.2 / FG-B .10.3
VLAN gateways: shared 10.40.<VLAN>.1
HA heartbeat: ha1 + ha2; see C09 / C10

SERVICE / POLICY NOTES
No PBX, SSH or firewall GUI exposed to Internet
Portal terminates at PROXY 10.40.80.10/24
DMZ gateway: 10.40.80.1; restricted backend access
Partner allowlist: 203.0.113.64/28 (fictional)
Offsite replication uses VPN, with no public DNAT

NAT rows are proposed documentation, not a deployment action. No credentials or customer-derived service rules are included.